



SCHOOL OF

INFORMATICS, COMPUTING, AND ENGINEERING

ENGR E210 – Engineering Cyber-Physical Systems, Section 7032

ENGR E599 – Digital Design (Topics in ISE), Section 9158

CSCI B441 – Digital Design, Section 7520

CSCI B541 Hardware System Design I, Section 7518

Syllabus

Prerequisites:

Prerequisites for E210 – Engineering Cyber-Physical Systems:

ENGR E110 Engineering Computing Architectures

ENGR E201 Computer Systems Engineering

Prerequisite for B441 – Digital Design:

CSCI-C 335 Computer Structures

Course Description:

This course introduces core topics in cyber-physical and digital systems. The first part of the course lays out a foundation for modern cyber-physical system design. These topics include embedded systems, issues of real-time processing, and sensor mechanisms and control algorithms. Students will study applications of these elements on the Internet of Things and Robotics.

The second part of the course covers digital system design using hardware description languages. We start with basics of digital electronics and learn how digital gates are used to build large digital systems. We will practice modern digital system design by using state of the art software tools and implementation of the digital systems on a programmable hardware platform.

Meeting Dates:

Lectures:	Tuesdays. Thursdays	9:35 AM–10:50 AM	Luddy Hall (IF) 1106
Lab (E210, 31810):	Wednesdays	2:05 PM–4:05 PM	Luddy Hall (IF) 4111
Lab (E210, 7033):	Wednesdays	4:20 PM–6:15 PM	Luddy Hall (IF) 4111
Lab (E599, 9159):	Wednesdays	4:20 PM–6:15 PM	Luddy Hall (IF) 4111
Lab (B441, 7521):	Wednesdays	4:20 PM–6:15 PM	Luddy Hall (IF) 4111
Lab (B541, 7519):	Wednesdays	4:20 PM–6:15 PM	Luddy Hall (IF) 4111

Course delivery method:

In person.

Learning Outcomes:

After taking this course, you will understand the underlying concepts of cyber-physical and digital systems you will be able to achieve the following objectives:

- Design and create a networked sensor system.
- Apply communication networks using UART, I2C, and SPI.
- Develop Python programs as part of a cyber-physical system.
- Develop System Verilog in the context of a larger system.
- Use a hardware description language along with an appropriate methodology to design and implement digital systems on a programmable hardware platform.
- Demonstrate the ability to utilize a version control system.
- Perform effectively in a small design team.

Assessments:

There will be lab projects and two exams.

Grade percentages:

Lab projects	50 %
Exam 1	25 %
Exam 2	25 %

Major Due Dates:

Note: These dates are subject to change with adequate notification via canvas announcements.

The labs are one-week or two-weeks. Lab reports are due Friday next week.

Exam 1:	February 24
Exam 2:	April 23

Required material:

We will cover selected chapters from the textbooks:

- *Introduction to Embedded Systems – A Cyber – Physical Systems Approach*, Second Edition, Edward Ashford Lee, Sanjit Arunkumar Seshia, The MIT Press, ISBN 978-1-312-42740-2.
- *Digital Design: An Embedded Systems Approach Using Verilog*, Peter J. Ashenden, Elsevier, Morgan Kaufman, ISBN-13: 978-012369527.
Electronic copies are available through the [IU library](#).

Reference materials:

- *ProGit book*, Scott Chacon and Ben Straub, available at:
<https://git-scm.com/book/en/v2>
- Web sites with data sheets will be provided in the lecture slides